

Practical No: 06

Write a program to implement POS tagging.

```
In [1]: print("125 Prosenjeet Shil")
```

125 Prosenjeet Shil

```
In [3]: import nltk
nltk.download('punkt')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\prose\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Error loading punkt_tab: Package 'punkt_tab' not found in
[nltk_data]   index
```

Out[3]: False

```
In [5]: from nltk import pos_tag

text="I enjoy coding and solving complex programming challenges."

tokens = text.split()
print ("Tokenized words: \n",tokens)
```

Tokenized words:

['I', 'enjoy', 'coding', 'and', 'solving', 'complex', 'programming', 'challenges.']

```
In [9]: nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]   C:\Users\prose\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping taggers\averaged_perceptron_tagger_eng.zip.
```

Out[9]: True

```
In [11]: tagged_words = pos_tag(tokens)
print("\nPOS tagged words : \n", tagged_words)
```

POS tagged words :

[('I', 'PRP'), ('enjoy', 'VBP'), ('coding', 'VBG'), ('and', 'CC'), ('solving', 'VBG'), ('complex', 'JJ'), ('programming', 'NN'), ('challenges.', 'NN')]

```
In [15]: for word, tag in tagged_words:
    print(f"{word} : {tag}")

tree_obj = nltk.Tree("Sentence", [(word, tag) for word, tag in tagged_words])
tree_obj.pretty_print()
```

I : PRP
enjoy : VBP
coding : VBG
and : CC
solving : VBG
complex : JJ
programming : NN
challenges. : NN

Sentence

I/PRP enjoy/VBP coding/VBG and/CC solving/VBG complex/JJ programming/NN challenges./NN